

Executive Summary

Project: Retail Sales Analysis - SQL & Excel

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Business Objective

Analyse retail sales performance using PostgreSQL and Microsoft Excel to help management monitor revenue, profitability, customer behaviour, product performance, regional performance and sales trends.

Analysis Scope

Sales Performance	Sales, profit, margin, orders and average order value
Product Analysis	Top products, loss-making products and discount impact
Customer Analysis	Segments, top customers, value groups and retention opportunities
Regional Analysis	Region, state and city-level profitability and loss-making markets
Time Analysis	Monthly trends, yearly growth, cumulative sales and rolling averages

Key Findings

- Total Sales reached **\$2.33M**.
- Total Profit reached **\$292K**, with an overall profit margin of **12.56%**.
- Technology generated the highest sales among product categories.
- Consumer customers contributed the largest share of revenue.
- The West and East regions generated the strongest profit performance.
- Several products and geographic areas showed negative profitability and require review.
- Monthly sales showed growth over time, with visible seasonality and variation.

Recommendations

- **Product Strategy:** Prioritise high-performing categories such as Technology while reviewing low-margin and loss-making products.
- **Regional Performance:** Investigate states and cities with negative profit to identify pricing, discount or operational issues.
- **Customer Strategy:** Target high-value and frequent customers for retention and growth opportunities.
- **Discount Management:** Review high-discount transactions and product groups to reduce margin pressure.
- **Planning:** Monitor monthly sales trends to support forecasting, inventory planning and campaign timing.

Business Value

This project demonstrates an end-to-end business analysis workflow: importing retail data into PostgreSQL, writing SQL analysis modules, exporting clean outputs, creating an Excel KPI report and translating results into management recommendations.

Tools Used: PostgreSQL, pgAdmin 4, Microsoft Excel, GitHub